

PCAccess — FileAccessAgent & Comparison Guide

Host Workstation Sharing Agent & Dual-Agent Architectural Blueprint

HOST WORKSTATION

SIGNALR WEBSOCKET

WINDOWS SERVICE

COMPARATIVE ANALYSIS

1. Introduction: FileAccessAgent Kya Hai?

FileAccessAgent ek secure background agent hai jo us **Host Computer / Server** par install hota hai jahan ke physical folders (e.g. D:\CompanyFiles) ko doosre users ke sath share karna hota hai.

Key Architecture: Yeh agent TeamViewer / AnyDesk ki tarah outbound WebSocket (SignalR) connection banata hai. Iske liye router me **koi port forwarding ya public IP kholne ki zaroorat nahi hoti**.

2. FileAccessAgent Pairing & Setup Workflow

- Run Host Agent:** FileAccessAgent\run.bat chalayein ya Windows Service install karein.
- Generate 6-Digit Pairing Code:** Agent launch hone par screen par ek temporary 6-digit code (e.g. 482-910) generate hoga jo 15 minute ke liye valid rehta hai.
- Pair via Web Portal:** Admin web portal me jakar *Connected Devices* > *Pair Device* me ye 6-digit code enter karega.
- Device Becomes ONLINE:** Device verify hote hi server par permanent token save ho jata hai aur device ka status live **ONLINE** badge dikhane lagta hai.
- Assign Folders:** Admin ab is machine ke kisi bhi local folder ko select karke specific users (ya sabhi) ko share kar sakta hai.

3. Windows Service Controls (Host PC par)

Host machine par 24x7 background me chalane ke liye pre-configured batch scripts maujood hain:

Batch Script	Equivalent Command	Action Performed
install-service.bat	FileAccessAgent.exe --install	Agent ko official Windows Service ke roop me register karta hai (Auto-starts on PC boot).
start-service.bat	FileAccessAgent.exe --start	Installed Windows service ko start karta hai.
stop-service.bat	FileAccessAgent.exe --stop	Running service ko safely halt karta hai.
check-status.bat	FileAccessAgent.exe --status	Service status (Running/Stopped) aur recent activity logs display karta hai.

Batch Script	Equivalent Command	Action Performed
uninstall-service.bat	FileAccessAgent.exe --uninstall	Windows Service registration ko system se cleanly remove karta hai.

4. Deep Comparison: UserLoginAgent vs FileAccessAgent

Dono agents ka PCAccess ecosystem me alag-alag aur complementary role hai. Niche di gayi table dono ka complete difference aur relation explain karti hai:

Feature	FileAccessAgent (Host Agent)	UserLoginAgent (Client Agent)
Primary Role	Folder Provider / Host Apne PC ke physical folders ko network me share karta hai.	Folder Consumer / Client Shared folders ko apne PC par Virtual Drive R: bana kar access karta hai.
Kahan Install Hota Hai?	Us machine par jahan actual files/folders physically store hain (Office Server, Admin PC).	User / Employee ke personal laptop ya desktop par.
Authentication Method	Temporary 6-Digit Pairing Code (Web Admin approval ke sath).	Username / Email + Password login.
Windows Mount (R:)	Nahi karta (Sirf service / daemon ki tarah chalta hai).	Haan! Login hote hi virtual R: ("Remote Drive") mount karta hai.
Connection Technology	SignalR Hub WebSocket (Live bi-directional file streaming).	REST API Authentication + Win32 Virtual Drive Staging.
Mode of Operation	Background Windows Service (24x7 unattended).	Interactive Desktop Console Application.

5. Common Questions & Interoperability (Dono Milkar Kaise Kaam Karte Hain?)

Q1: Kya UserLoginAgent ko chalane ke liye FileAccessAgent ka live hona zaruri hai?

Nahi! UserLoginAgent bina FileAccessAgent ke bhi login karega aur user ke PC me R: Remote Drive create ho jayegi. Permitted folders ke naam aur structure user ko dikhenge.

Q2: Phir FileAccessAgent kab zaruri hai?

Jab user ko kisi remote PC ke folder ke andar ki actual real-time files browse karni ho ya download/upload karni ho, tab us specific host PC ka FileAccessAgent live/online hona zaruri hai.

Q3: Kya dono agent ek hi PC par chal sakte hain?

Haan! Agar aapka PC hi Host PC hai aur aap khud apne user account se R: drive mount karna chahte hain, to dono agents same PC par bina kisi conflict ke chal sakte hain.